



# AI-Based Customer Churn Prediction Using ANN

Resume-Worthy Mini Project (TensorFlow + Scikit-Learn)



## Objective

Build an Artificial Neural Network (ANN) to predict whether a telecom customer is likely to leave the company (Churn Prediction) based on customer usage and support history.



## Business Problem

Telecom companies lose customers due to poor service, high costs, and frequent complaints. The goal is to identify customers at risk of leaving so retention strategies can be applied in advance.



## Dataset

### Features

- **Tenure** (Months with company)
- **Monthly Charges**
- **Data Usage (GB)**
- **Support Tickets Raised**

### Target

- **Churn**
  - 0 = Customer Stays
  - 1 = Customer Leaves



## Technology Stack

- NumPy
- Pandas
- Scikit-Learn
- TensorFlow / Keras



## Project Workflow

### Step 1: Generate Dataset

Create 1000 customer records using NumPy and Pandas.

### Step 2: Data Preparation

- Create DataFrame
- Separate Features (X) and Target (y)

### Step 3: Train-Test Split

```
test_size = 0.2
```

```
random_state = 42
```

### Step 4: Feature Scaling

Apply:

```
StandardScaler()
```

```
..
```



## ANN Architecture

Input Layer (4 Features)

↓

Dense(16, ReLU)

↓

Dense(8, ReLU)

↓

Dense(1, Sigmoid)



## Model Configuration

```
optimizer = 'adam'
```

```
loss = 'binary_crossentropy'
```

```
metrics = ['accuracy']
```

Training Parameters:

```
epochs = 100
```

```
batch_size = 32
```

```
validation_split = 0.2
```



## Evaluation

Display:

- Accuracy
- Loss

- Model Summary

Predict churn status for a new customer and display the result.



## Visualizations

### Graph 1

Training Accuracy vs Validation Accuracy

### Graph 2

Training Loss vs Validation Loss



## Deliverables

Submit:

customer\_churn\_prediction.py

customer\_churn\_prediction.ipynb



## Resume Description

### AI-Based Customer Churn Prediction Using Artificial Neural Networks

- Developed an ANN model using TensorFlow/Keras to predict customer churn.
- Performed preprocessing, train-test splitting, and feature scaling using Scikit-Learn.
- Built a binary classification model with multiple hidden layers.
- Evaluated model performance using accuracy and loss metrics.
- Visualized model learning using training and validation curves.
- Solved a real-world business problem focused on customer retention and predictive analytics.



## Interview Questions

1. Why is this a classification problem?
2. Why is Sigmoid used in the output layer?
3. What is Binary Cross Entropy?
4. Why do we perform feature scaling?
5. What is the purpose of validation data?
6. What is overfitting?
7. Why is customer churn prediction important?
8. How does ANN learn from data?



## Outcome

A professional ANN-based prediction project suitable for fresher resumes targeting **AI Engineer, ML Engineer, Data Analyst, Data Science, and GenAI Associate** roles.